Forecastability of Accessibility Deterioration Under Recoverability Constraints: A Minimal Computational Note

Toward an operational distinction between preserved mapping and dynamic routing

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Abstract

Many systems may remain structurally preserved while progressively reorganizing which future trajectories remain operationally accessible under repeated perturbation and recovery constraints. However, it remains unclear whether such deterioration in accessibility may become operationally forecastable prior to explicit collapse.

Here, we introduce a minimal computational framework exploring whether future deterioration in operational efficiency may become forecastable from internal recoverability state rather than perturbation geometry alone. Using a recoverability-constrained dynamical abstraction, we construct temporal operational datasets and evaluate predictive performance across multiple future horizons under repeated perturbation conditions.

The simulations suggest that future deterioration may become strongly forecastable from operational recoverability observables, whereas perturbation geometry alone exhibits substantially weaker predictive performance. Forecastability remains stable across future horizons and appears primarily organized around recoverability state variables, particularly mean recoverability state, rather than external perturbation descriptors. Importantly, randomized control tests reduce predictive performance toward chance levels, consistent with a non-trivial operational signal under the present abstraction.

The present work should be interpreted conservatively as an exploratory and falsification-oriented computational abstraction rather than a fitted mechanistic model of real systems. More broadly, the results may be consistent with an operational distinction between preserved structural mapping and dynamically reorganized routing, in which comparable scaffolds coexist with progressively changing reachable futures.

1. Introduction

When preserved structure does not imply preserved future accessibility.

Many dynamical systems may remain structurally intact while progressively reorganizing which futures remain operationally accessible. In such cases, preserved state-space availability or scaffold organization may not necessarily imply preserved reachability under repeated perturbation, recovery mismatch, accumulated stress, or path-dependent dynamics.

Operationally, this raises a distinction between what remains structurally available and what remains dynamically traversable in practice.

One possible way to frame this distinction is through the difference between mapping and routing.

Mapping refers to the preserved scaffold of available structure: the topology of states, interactions, or trajectories that formally remain present and structurally admissible. Routing, by contrast, refers to which trajectories remain dynamically traversable under operational constraints.

Under this interpretation, a system may preserve similar structural organization while progressively reorganizing which trajectories remain reachable, efficient, or recoverable.

Accordingly, identical perturbation burden does not necessarily imply identical accessible futures. Systems sharing similar perturbation geometry may nevertheless exhibit markedly different future accessibility depending on recoverability state, perturbation timing, accumulated perturbation history, or stochastic variability.

The present work explores whether such deterioration may become operationally forecastable prior to explicit collapse. Specifically, we ask whether future accessibility deterioration may emerge predictively from internal recoverability observables rather than perturbation geometry alone.

To investigate this question, we introduce a minimal computational forecasting framework designed as a falsifiable operational abstraction rather than a mechanistic model of any particular real system.

2. Minimal Operational Forecasting Framework

A recoverability-constrained computational abstraction.

To investigate whether future accessibility deterioration may become operationally forecastable, we introduce a minimal recoverability-constrained computational abstraction designed to simulate repeated perturbation, recovery, and accumulated operational degradation over time.

The framework represents a simplified accessibility lattice in which local recoverability and accumulated burden jointly shape operational traversability. Rather than modeling any specific biological, ecological, or engineered system, the objective is to evaluate whether minimal operational observables may contain predictive information about future deterioration under repeated perturbation and recovery constraints.

At each time step, the system evolves according to a small set of operational variables describing recoverability, perturbation exposure, and accumulated burden. The model tracks both instantaneous operational state and its temporal evolution under repeated perturbation schedules.

2.1 Operational observables

The forecasting framework evaluates a small set of operational observables:

- Mean recoverability state (mean_r): average recoverability across the system, interpreted as an operational proxy for current accessibility organization.
- Minimum recoverability (min_r): lowest local recoverability observed in the system.
- Perturbation interval: spacing between repeated perturbation events.
- Perturbation density: effective perturbation frequency.
- Recovery rate (β): parameter governing system recovery dynamics.
- Recovery constraint (γ): parameter regulating degradation pressure on recoverability.
- Damage decay: relaxation rate of accumulated perturbation burden.

Together, these variables provide a minimal operational description of system organization under perturbation–recovery dynamics.

2.2 Forecasting target

Rather than predicting explicit collapse, the framework evaluates whether future deterioration becomes operationally forecastable from present system organization.

A binary forecasting target, future deterioration, is defined according to whether operational efficiency decreases over a subsequent observation window:

operational state at time t → deterioration at $t + \Delta t$

This formulation treats deterioration as a future accessibility problem rather than a structural failure event. The objective is not to predict catastrophic collapse itself, but to determine whether measurable operational deterioration may become forecastable before explicit breakdown occurs.

2.3 Forecast horizon and falsation strategy

Forecastability is evaluated across multiple future horizons (Δt = 10–40) to determine whether predictive structure persists beyond immediate next-step dynamics.

Several falsation-oriented controls are additionally introduced:

- Geometry-only forecasting, testing whether perturbation structure alone carries predictive information.
- Minimal operational forecasting, testing whether internal operational observables recover predictive signal.
- Shuffle-target controls, testing whether predictive performance collapses under randomized labels.

Under this setup, predictive performance is interpreted conservatively: not as evidence of mechanism, but as evidence that present operational organization may contain predictive information regarding future accessibility deterioration.

3. Results

Operational forecastability under recoverability constraints

3.1 Forecastability from operational state versus perturbation geometry

To evaluate whether future deterioration may become operationally forecastable, predictive performance was compared across progressively constrained forecasting configurations.

The first configuration evaluates perturbation geometry alone, using perturbation interval, density, and recovery-related parameters without access to present operational recoverability state. A second configuration evaluates recoverability state alone using mean recoverability (mean_r). Additional models examine a minimal operational formulation and the complete feature set. Finally, shuffle-label controls are used to test whether predictive performance collapses under destroyed predictive structure.

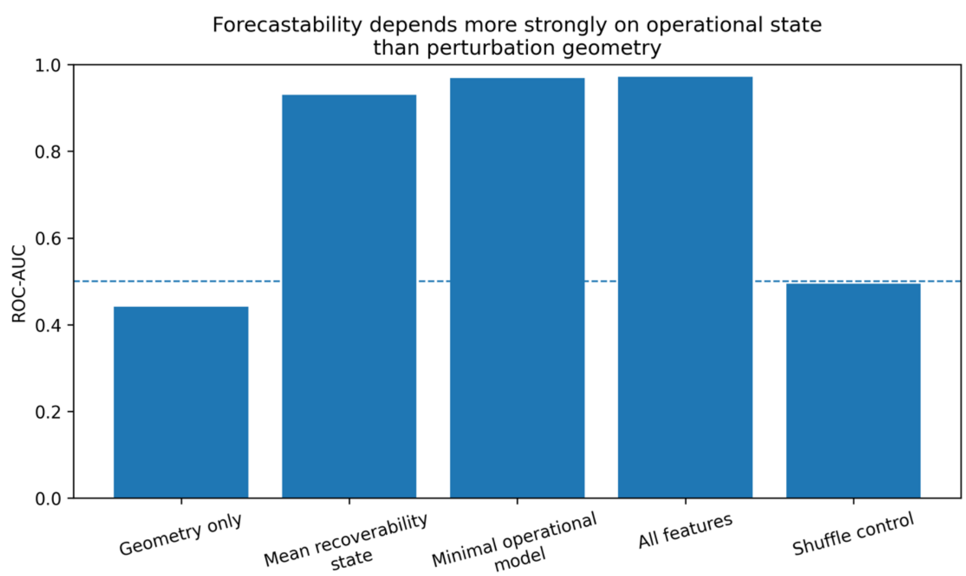

Figure 1. Forecastability comparison across operational and structural forecasting conditions. *Predictive performance (ROC-AUC) across forecasting configurations. Perturbation geometry alone exhibits weak predictive performance, whereas present operational recoverability state strongly forecasts future deterioration. Minimal operational observables recover near-maximal predictive signal, while shuffle-target controls collapse toward chance levels, consistent with non-trivial forecastability.*

Several observations emerge under the present operational formulation.

First, perturbation geometry alone exhibits weak predictive discrimination (ROC-AUC ≈ 0.44), suggesting that perturbation schedule by itself may contain limited information regarding future deterioration.

Second, recoverability state alone achieves substantially stronger predictive performance (ROC-AUC ≈ 0.93), consistent with the possibility that present operational organization already compresses information relevant to future accessibility deterioration.

Third, a minimal operational formulation combining recoverability and perturbation-related observables recovers near-maximal predictive performance (ROC-AUC ≈ 0.97), while inclusion of the full feature set provides only marginal additional improvement.

Importantly, shuffle-label controls reduce predictive performance toward chance levels (ROC-AUC ≈ 0.50), consistent with the presence of non-trivial predictive structure under the present abstraction rather than trivial statistical leakage.

Taken together, these observations are consistent with the possibility that future deterioration may become operationally forecastable primarily through present recoverability organization rather than perturbation geometry alone.

3.2 Forecastability persists across future horizons

A second question concerns temporal persistence: does predictive structure decay rapidly, or remain informative across future forecasting horizons?

To evaluate this, predictive performance was examined across multiple forecasting horizons (Δt = 10–40).

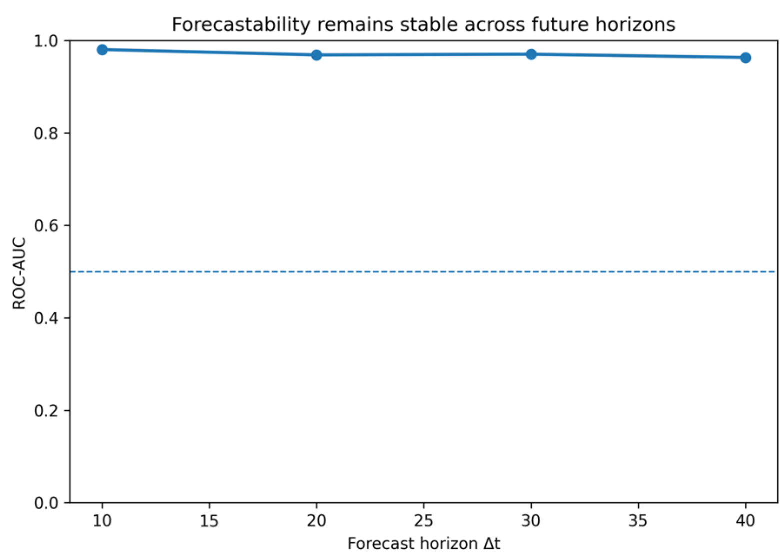

Figure 2. Forecastability across future horizons. *Predictive performance remains consistently high across multiple future forecasting horizons (Δt = 10–40), suggesting that operational forecastability persists beyond immediate next-step deterioration within the present computational abstraction*

Forecastability remains consistently high across all tested horizons (ROC-AUC ≈ 0.96–0.98), suggesting that predictive information is not restricted to immediate next-step deterioration.

Under the present abstraction, this observation is consistent with persistent operational organization, in which present system state continues to constrain future accessibility beyond local perturbation events.

Importantly, this should not be interpreted as evidence of mechanistic memory in any specific real system, but rather as an operational observation regarding temporal persistence within the computational formulation.

3.3 Predictive contribution of operational observables

To investigate which operational observables contribute most strongly to forecastability, permutation importance analysis was performed on the minimal operational formulation.

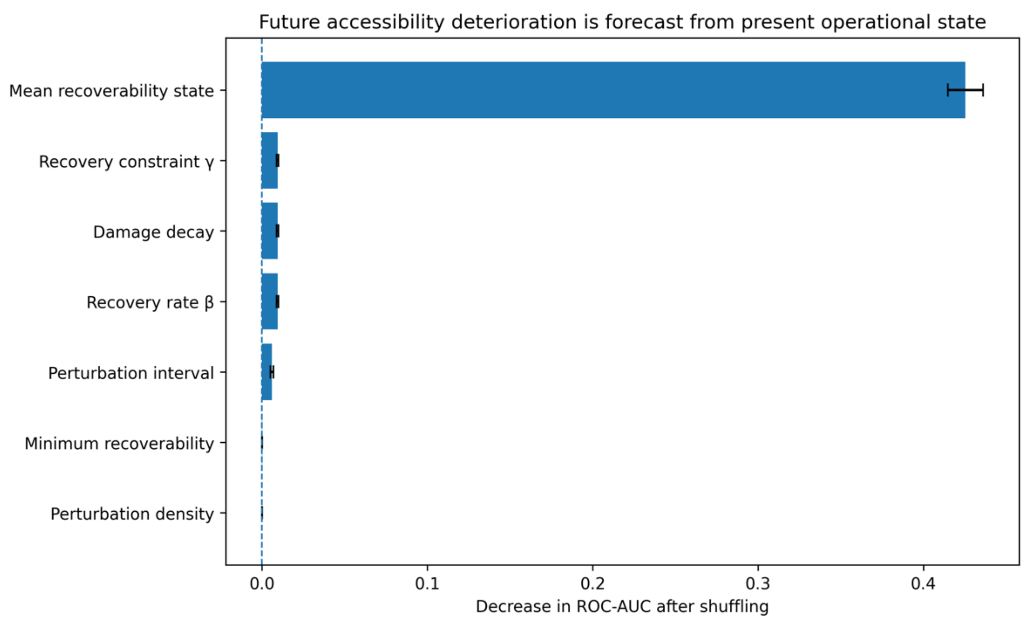

Figure 3. Permutation importance of operational observables. *Mean recoverability state (mean_r) provides the dominant predictive contribution, while perturbation timing and recovery-related parameters contribute smaller but non-zero predictive signal. Results are consistent with predictive information becoming operationally compressed into present recoverability state.*

Mean recoverability state (mean_r) provides the dominant predictive contribution, whereas perturbation timing and recovery-related parameters contribute substantially smaller but non-zero signal.

This observation is consistent with the possibility that predictive information becomes operationally compressed into present recoverability organization rather than remaining explicitly encoded in perturbation geometry itself.

In other words, future deterioration appears weakly forecastable from perturbation schedule alone, but becomes substantially more forecastable once perturbation history is reorganized into present operational recoverability state.

This observation motivates an operational distinction between preserved mapping and realized routing, developed further in the interpretation section.

3.4 Predictive performance characteristics

For completeness, predictive discrimination was additionally evaluated using receiver operating characteristic (ROC) analysis.

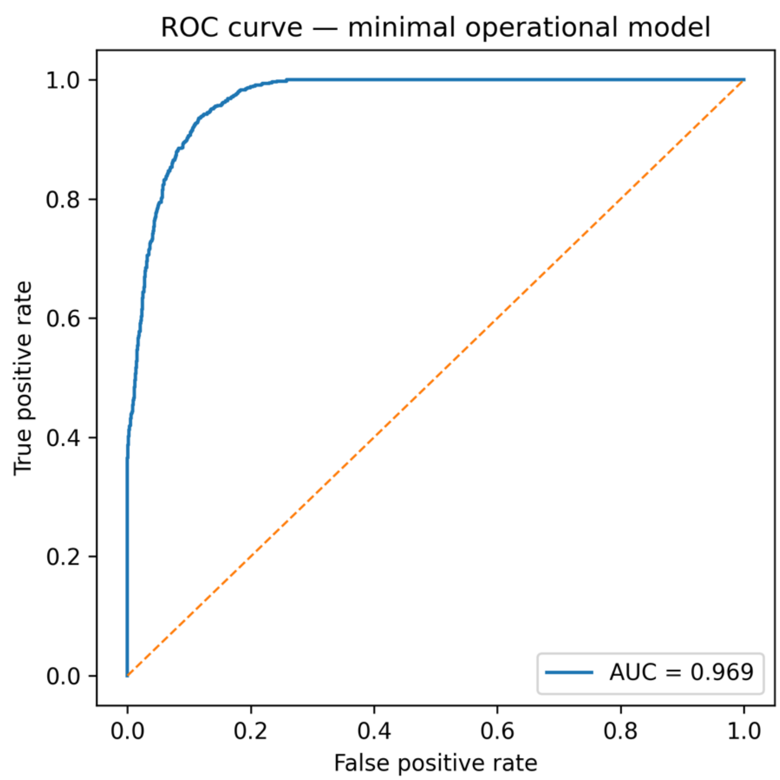

Figure 4. Receiver operating characteristic (ROC) curve for the minimal operational model. *Predictive discrimination remains consistent with strong separation between deterioration and non-deterioration trajectories under the operational forecasting formulation.*

The resulting discrimination profile remains consistent with substantial separability between deterioration and non-deterioration trajectories under the operational forecasting formulation.

Under the present abstraction, this result supports the interpretation that predictive signal remains operationally detectable rather than indistinguishable from random discrimination.

4. Interpretation

Preserved mapping does not necessarily imply preserved routing

The present results are consistent with an operational distinction between preserved mapping and realized routing under recoverability constraints.

Within this interpretation, mapping refers to preserved scaffold organization: the structural topology of states, interactions, or formally admissible trajectories that remain present within a system. Mapping therefore describes what remains structurally available or potentially reachable.

Routing, by contrast, refers to which trajectories remain dynamically traversable in practice under operational constraints. Routing therefore concerns accessibility, recoverability, and the evolving subset of futures that remain operationally reachable as perturbation history accumulates.

Under this framing, future deterioration appears only weakly forecastable from perturbation geometry alone, but becomes substantially more forecastable once perturbation history is reorganized into present operational recoverability state.

Operationally, this distinction suggests that systems may preserve similar structural organization while progressively reorganizing which futures remain accessible. In this sense, identical perturbation burden does not necessarily imply identical accessible futures.

A preserved scaffold may therefore coexist with progressively reorganized routing capacity.

Importantly, this interpretation should not be understood mechanistically or universally. The present framework does not claim that real biological, ecological, technological, or cognitive systems necessarily obey this organization. Rather, the results are consistent with the possibility that a recoverability-constrained operational abstraction may generate conditions under which future accessibility deterioration becomes forecastable despite preserved structural organization.

One possible interpretation is that present operational state behaves as an operational compression of accumulated perturbation history. Under repeated perturbation and recovery mismatch, predictive structure may progressively reorganize into present state variables, making future deterioration increasingly forecastable before explicit collapse.

Under this interpretation, deterioration becomes less a problem of structural disappearance and more a problem of changing reachability.

In minimal form, the present results are consistent with the possibility that preserved mapping does not necessarily imply preserved routing.

5. Limitations

Interpretational boundaries

Several limitations should be emphasized.

First, the present framework should be interpreted as an exploratory computational abstraction rather than a fitted mechanistic model of any particular biological, ecological, technological, or cognitive system. The operational variables are intentionally minimal and designed to investigate forecastability under recoverability constraints rather than reproduce real-world mechanisms.

Second, predictive performance should not be interpreted as evidence of universal predictive organization. Rather, the present simulations are consistent with the possibility that, under the assumptions of this minimal operational formulation, predictive structure may emerge once perturbation history becomes reorganized into present recoverability state.

Third, the operational distinction between mapping and routing should be understood as a conceptual interpretation of the present computational results rather than an ontological claim regarding system organization. The framework merely suggests that systems preserving comparable structural organization may nevertheless differ in which future trajectories remain dynamically traversable.

Fourth, no claim is made regarding direct empirical applicability. Additional work would be required to determine whether analogous accessibility forecasting signals emerge in experimentally measurable systems involving repeated perturbation, recovery mismatch, hysteresis-like dynamics, resilience loss, or progressive operational degradation.

Finally, the forecasting target itself emerges from the same operational dynamics generating recoverability state. Accordingly, predictive success should not be interpreted as evidence of externally validated predictability, but rather as evidence that recoverability-constrained operational organization may, in principle, contain information relevant to future deterioration under the assumptions of the present abstraction.

6. Conclusion

Toward operational forecastability of accessibility deterioration

This work explored whether future accessibility deterioration may become operationally forecastable under repeated perturbation and recoverability constraints.

Within a minimal computational abstraction, perturbation geometry alone exhibited weak predictive performance, whereas operational recoverability observables retained strong predictive signal across multiple future horizons. Predictive performance remained stable across temporal horizon variation and collapsed toward chance levels under randomized controls, consistent with non-trivial operational structure within the forecasting formulation.

The dominant predictive contribution emerged from present recoverability state, particularly mean recoverability, suggesting that accumulated perturbation history may become operationally compressed into current system organization.

More broadly, the results are consistent with a distinction between preserved mapping and realized routing: systems may preserve comparable structural scaffolds while progressively reorganizing which future trajectories remain operationally reachable under operational constraints.

The present work should be interpreted conservatively: as an exploratory and falsifiable computational abstraction investigating whether future accessibility deterioration may become forecastable prior to explicit collapse, rather than as evidence for a mechanistic or universal principle across real systems.

7. Reproducibility

All simulations, forecasting analyses, figures, exported datasets, and supporting materials associated with this computational note are available through the accompanying reproducible repository and executable notebook.

The repository includes:

- Executable forecasting notebook
- Exported simulation datasets
- Forecasting performance metrics
- Permutation importance analyses
- Forecast horizon robustness analyses
- Publication figures used in this note

The accompanying notebook is intended to support full reproduction, falsification, modification, and extension of the present computational abstraction under alternative operational assumptions.

8. Acknowledgements

This work was developed through an exploratory human–AI scientific workflow using large language model assistance for computational structuring, implementation support, manuscript refinement, and reproducibility organization.

The computational assistance provided during development supported implementation, writing iteration, falsification-oriented organization, and repository preparation. Conceptual framing, interpretation, scientific positioning, and responsibility for the present work remain with the author.

The present study is intended as part of an open, reproducible, and iterative scientific process in which computational assistance may contribute to accelerating exploratory hypothesis generation, testing, and refinement under explicit interpretational constraints.

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